Society of Petroleum Engineers

SPE-229989-MS

Transforming Challenges into Opportunities - Field-Scale Success Case Studies and Lessons Learned from Polymer Flooding Using Produced Water in Harsh Environments of Several Complex Heterogeneous Carbonate and Sandstone Reservoirs of Kuwait

M. Y. Khan, H. F. Abdulraheem, S. K. Eadulapally, and B. Al-Otaibi, Kuwait Oil Company, Kuwait

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

The field success of effective polymer flooding hinges on in-situ oil viscosity, mobile oil saturation, reservoir temperature, conformance, injectivity, shear rate, brine chemistry (including salinity, iron, and divalent cations), polymer concentration, adsorption, and polymer characteristics. This paper presents a series of novel field-scale case studies of polymer flooding using high total dissolved solids (TDS) produced water injection, highlighting lessons learned from lab-to-successful field implementation under harsh environments of complex heterogeneous carbonate and sandstone reservoirs.

Technically innovative and systematic workflows were developed for evaluation and deployment of polymer flooding under harsh reservoir conditions. Stable synthetic polymers were rigorously screened via laboratory and core flooding tests using high-TDS water and harsh reservoir environments. History-matched, high-resolution simulations were performed to design pre-pilot injection strategies. Long-term-polymer-injectivity-tests (LTPIT) were executed in multiple reservoirs using a single well to assess polymer techno-commercial viability (rates/throughput), matrix injectivity, in-situ performance and advanced surveillance. Lessons learned enabled proactive mitigation or the transfer of challenges into opportunities. Post-LTPIT calibrated models' simulations were performed to design commercial polymer flooding and economic assessments across diverse reservoirs.

Screening and laboratory tests revealed that synthetic polymers composed of 20–50% ATBS and 5–18 million Daltons molecular weights exhibit chemical, mechanical, and thermal stability under harsh environments up to 270,000 ppm of TDS, 40,000 ppm of hardness and 210°F temperature. Core flood tests using high-TDS injection water demonstrated these polymers' potential to increase oil recovery and accelerate oil production in heterogeneous carbonate and sandstone reservoirs with 2–100 cp oil viscosities. Field trials through LTPITs established polymer long-term injectivity under matrix conditions, techno-commercial viability (injection rates/throughput), in-situ polymeric rheological parameters (viscosity/concentration/shear rate), residual resistance factor (RRF) and resistance factor (RF). Lessons learned from each LTPIT successfully addressed to convert challenges into opportunities in subsequent pilots, such as reservoir heterogeneity, low injectivity, scaling, rheology deviations, well placement, pilot duration, pilot result upscaling, fracture pressure, downhole pressure, and RF/RRF interpretation. Data from LTPITs were used to calibrate high-resolution reservoir simulations, enabling successfully upscaling for full-field commercial deployment. which demonstrated improved oil recovery, accelerated oil production, and reduced water cut compared to waterflooding. Economic analysis across diverse reservoir settings showed favorable unit technical cost (UTC) and increased net present value (NPV), confirming the techno-commercial viability of polymer flooding in very harsh environments. These case studies reveal the successful polymer flooding using high-TDS produced water in the harsh environment of geologically complex, heterogeneous carbonate and sandstone reservoirs. The findings highlight the lessons learned from lab-to-field applications as tailor-made polymers' selection from labs, overcoming operational/ geological challenges, transforming challenges into opportunities, and leveraging surveillance for improved implementation. The findings offer actionable insights that pave the way for field applications of polymer flooding in similarly challenging reservoir environments.

Keywords: Polymer flooding, incremental oil production, long term polymer injectivity, harsh environment, High TDS produced water

Introduction

Polymer flooding is considered as the most widely utilized and advanced proven technique in enhanced oil recovery (EOR)/ Improved Oil Recovery (IOR) technology. The main goal of polymer flooding is to mainly enhance sweeping by raising the viscosity of the injected aqueous phase (Sorbie 1991). Since the 1950s, numerous laboratory studies, pilot field deployments, and full-scale field projects have been conducted (Standnes and Skjevrak, 2014). Synthetic polymers, particularly hydrolyzed polyacrylamides (HPAM), are chosen due to their cost-effectiveness and higher biodegradation resistance (Hassan et al. 2022). HPAM is produced by polymerizing acrylamide monomers into long-chain polymers. The process of hydrolyzing polyacrylamide introduces a negatively charged carboxylate group, replacing some of the amide groups present in the backbone chain. The viscosity of HPAM arises from the anionic repulsion of acrylate moieties, which elongate the molecular structure (Lake et al. 2014). When the concentration of brine divalent cations (Ca^{++} /Mg^{++}) rises, these cations attract the negatively charged carboxylate groups, leading to a decrease in repulsive intramolecular forces and a decrease in polymer viscosity (Levitt and Pope 2008). Additionally, polymer concentration, shear rate, temperature and molecular weight are crucial factors influencing polymer viscosity (Jouenne and Levache 2020). Although HPAM is a widely used synthetic polymer in most polymer injection projects, it undergoes excessive thermal hydrolysis at high temperatures and tends to precipitate in the presence of divalent ions (Ca^{++} /Mg^{++}). Extensive efforts have been conducted to enhance oil recovery using synthetic polymers, and it has been proven that incorporating hydrophobic groups into the polymer backbone can significantly improve surface and interfacial activities. The introduction of sulfonamide, sulfonate groups and aromatic rings has been shown to enhance thermal stability and salt tolerance. The selection of an appropriate reservoir for polymer injection depends on various reservoir properties, including permeability, thickness, temperature, oil saturation, oil viscosity, brine composition and rock composition. Traditionally, polymer injections are recommended for sandstone formations with conditions of low temperature (<70°C), high permeability (>10 mD) and oil viscosity limit (<150 cP). Since polymer flooding primarily aims to enhance oil recovery through improved sweeping efficiency, oil saturation greater than 40% is considered necessary for economic viability. However, recent advancements in polymer flooding have expanded the applicability of polymers to challenging reservoir conditions, including temperatures above 70°C and total dissolved solids (TDS) levels exceeding 100,000 ppm (Hassan et al., 2022; Seright et al., 2021; Dupuis et al., 2017).

In such extreme conditions, evaluating polymer stability is a critical factor. At temperatures above 60°C, chemical degradation can lead to polymer precipitation, particularly in the presence of divalent ions in the brine. Chemical degradation primarily arises from the hydrolysis of the polymer's hydroxyl hanging radicals (NH_2, O-). The rate of hydrolysis increases with temperature and pH, which can reduce polymer viscosity and stability. Additionally, thermal stability can significantly affect polymer viscosity under harsh conditions, especially when oxygen, iron, or polymer impurities are present. For example, when oxygen is present, the half-life of polymer thermal stability can decline dramatically from 117 days to just 2 hours as the temperature rises from 50°C to 90°C. Oxygen can react with ferrous iron or polymer impurities, leading to significant degradation. Oxidation tends to occur more frequently in laboratory settings and near the wellbore, while deeper in the reservoir, a reducing environment is typically found. Any gas leakage during improper lab preparation could result in a misinterpretation of polymer stability for field applications. In the absence of ferric, divalent ions and, the polymer can hold at least 50% of its original viscosity for more than 8 years, even at elevated temperatures exceeding 90°C. Most documented polymer field projects have utilized oxygen scavengers and avoided using steel pipelines to minimize oxygen-iron reactions. However, oxygen scavengers can raise project costs and potentially accelerate microbial degradation. In conclusion, polymer stability is affected by various factors, including—but not limited to—oxygen, ferric, salinity, water hardness, temperature, pH, residuals, polymer structure and its molecular weight.

Kuwait has set an ambitious target for EOR, aiming for it to contribute approximately 10% of Kuwait Oil Company's total daily oil production. Among the various EOR techniques, polymer flooding has emerged as a key technology, with recent field tests demonstrating its effectiveness in improving oil recovery. However, implementing polymer flooding presents two critical challenges:

Existing polymers, particularly HPAM, undergo thermal degradation and viscosity loss in reservoirs with high temperatures, high salinity, and elevated concentrations of divalent ions. Thermally stable polymers are essential to optimize polymer flooding performance and enhance long-term project sustainability. The effectiveness of polymer flooding is contingent on the polymer's ability to maintain its viscosity over the extended travel time between injection and production wells. This requires proper handling to reduce impurities, oxygen and iron contamination, as well as a comprehensive evaluation of polymer thermal stability to ensure its performance under in situ reservoir conditions.

These challenges can address through laboratory and field implementation of polymer flooding for different type of reservoirs, which provides technical and financial advantages for Kuwait's field development plans and ensures the long-term success of polymer-based EOR strategies. Therefore, an extensive laboratory evaluation carried out to select suitable polymer for each reservoir keeping in mind their harsh environment as well injection water source as produced/effluent water. And systematic workflow from lab testing to field deployment was developed, using stable synthetic polymers {15–50% Acrylamide Tertiary Butyl Sulfonic Acid (ATBS}, 8–18 million Daltons) proven effective under high salinity (up to 270,000 ppm TDS), hardness (40,000 ppm) and temperatures up to 210°F. Long-Term Polymer Injectivity Test (LTPIT) pilot field trial planned for injectivity performance and economic viability. The pre-modeling was carried out to design injection sequence before field trail. Then LTPIT tests were performed in the field of different reservoirs with the objectives below:

- Establish polymer-flooding techno-commercial viability: Injection rates / throughput
- Establish polymer injectivity under matrix conditions
- Estimate residual resistance factor (RRF) and resistance factor (RF)
- Estimate in-situ polymeric rheological parameters: Viscosity/concentration/shear rate
- Pilot duration: Complete pilot within 6 months to meet the above pilot objectives
- Confirm functionality of effluent water treatment and polymer mixing/injection units
- Predict polymer response, oil peak and injectivity in different rock types: If possible

Further the simulation model of each reservoir was calibrated with acquired pilot data and history match the pilot to upscale these results for phase commercial polymer flooding development in each reservoir. The techno-economic oil recovery of polymer flood over water flood were predicted for each reservoir. This study showcases successful field-scale polymer flooding using high-TDS produced water in complex, harsh carbonate and sandstone reservoirs. Calibrated simulations guided pilot design and full-field development, resulting in higher oil recovery, faster production and lower water-cut. The case studies provide practical insights into polymer selection, overcoming operational challenges, and ensuring cost-effective polymer flooding implementation in the harsh reservoir conditions.

Case studies' reservoirs description

The choice of a suitable reservoir for polymer injections is contingent on various reservoir properties, including permeability, thickness, temperature, oil saturation, oil viscosity, brine composition, and rock composition etc. Therefore, extensive evaluations were conducted to identify the most appropriate candidates for field trials and initial commercial deployment of polymer flooding. A comprehensive laboratory and field assessment of LTPIT was carried out across multiple reservoirs to confirm long term polymer injectivity in the field before phase commercial polymer flooding development. As part of the EOR/IOR Screening, nine key reservoirs were selected as pilots and phased commercial polymer flooding implementation based on their rock and fluid compatibility with polymer injection. Influence of produced polymer on production facilities was identified from laboratory analysis. (Al-Mayyan et al., 2023; Dupuis et al., 2024). The Kuwaiti reservoirs where polymer long term injectivity was tested and commercial development phases were carried out are illustrated in Fig. 1 as follows:

North Kuwait: Sabriyah Upper Burgan (SAUB) and Raudhatain Upper Burgan (RAUB), both conventional sandstone reservoirs, along with Umm Niqa Lower Fars (UNLF), a heavy oil reservoir characterized by unconsolidated sandstone.

West Kuwait: Umm Gudair Minagish Oolite (UGMO), a carbonate reservoir, and Minagish Wara Burgan (MN-WABG), a sandstone reservoir.

South and East Kuwait: Four regions of the Greater Burgan Field as Western, Eastern, Northern and Central (known as 3SU) Wara Burgan Reservoir—comprising extensive sandstone formations.

The selected reservoirs span a wide range of reservoir conditions, as detailed in Table 1. These include permeability ranging from 100 to 1,000 mD, formation temperatures between 37°C and 83°C, in-situ oil viscosities from 1 to 100 cP and formation water TDS values ranging from 89,000 to 241,000 ppm. The presence of divalent cations (Ca^{++} /Mg^{++}) also varies significantly, from 11,589 to 31,693 ppm. These findings confirm that Kuwait's reservoirs present diverse but technically manageable challenges for polymer flooding, paving the way for full-scale EOR deployment under a variety of geological and fluid conditions.

Figure 1—Locations of selected reservoir for polymer flooding in state of Kuwait

Table 1—Description of rock and fluid properties of North, West and South-East Kuwait reservoirs

Location	North Kuwait			West Kuwait		South & East Kuwait			
Case Study	Case Study 1	Case Study 2	Case Study 3	Case Study 4	Case Study 5	Case Study 6	Case Study 7	Case Study 8	Case Study 9
Reservoir name	SAUB	RAUB	UNLF Heavy Oil	UGMO	MN-WABG	Eastern Wara Burgan	Western Wara Burgan	Northern Wara Burgan	3SU Reservoir
Formation	Sandstone	Sandstone	Unconsolidated Sandstone	Carbonate	Sandstone	Sandstone			
Reservoir heterogeneities	Faults, irregular developed channel sands	Faults, irregular developed channel sands	sallow reservoir with caprock integrity issue at higher pressure	Strong barrier, baffles	irregular developed channel sands	shale, Vclay, faults			
Reservoir depth (ft)	7395 to 7,800	7600	650	8000 to 9000	5400 to 5880				
Reservoir Thickness (ft)	46	160	16	266	30				
Reservoir Temperature (°C)	80	82	37	76 to 83	64	55	57	55	57
Average Porosity (%)	22	22	30	22	25	26	26	26	26
Permeability range mD)	40 to 4000	20 to 2000	20 to 20000	3 to 1500	5 to 1000	1 to 10000	1 to 10000	1 to 12500	1-12000
Average permeability (mD)	677	286	1000	245	300 to 700	1540	1569	263	528
Oil viscosity range (cp)	1 to 10	1 to 30	45 to 100	2 to 20	22 to 32	0.86 to 177	0.92 to 80	0.85 to 80	1.3 to 20
Oil API (degree)	18 to 31	18 to 30		14 to 30	18 to 31				
current water cut (%)	52	72	0	60	52	55	55	55	45
Initial reservoir pressure (psi)	3762	3855	330	4100	3762				
Saturation pressure (psi)	904	668 to 2229		600 to 1200	142				
Current reservoir pressure (psi)	2400	2445	330	2100	2400				
Formation TDS (ppm)	241000	241000	166721	215193	200000 to 2360000	210000	170000	127600	89000
Formation water divalent ions as Ca++ & Mg++ (ppm)	17985	19857	14300	31693	20000	11589	11589	11589	12216
Injected source water TDS (ppm)	193000	193000	123867	271613	243821	163000	163000	163000	163000
Injected source water divalent ions as Ca++ & Mg++ (ppm)	15200	16069	10800	56646	17319	12000	12000	12000	12000
Oil plateau decline from	2018	2017		2002	2000	no decline	no decline	2021	2015
Aquifer support	Weak aquifer from edge	Weak aquifer from eastern flank	weak aquifer from bottom	Strong aquifer from bottom and edge	Weak aquifer from edge with limited connectivity	Adequate aquifer support			

Methodology

This paper outlines the methodology adopted for the commercial implementation of polymer flooding. The process began with comprehensive laboratory testing to identify polymer formulations compatible with the reservoir and injection water conditions, which are characterized by high temperature, elevated salinity, and significant hardness. Following this, a pre-modelling exercise was conducted using a high-resolution simulation model, incorporating surveillance data collected during baseline water injection under the Long-Term Pilot Injection (LTPI) framework. Based on the simulation insights, a detailed injection sequence was designed, comprising baseline water injection, polymer injection, and subsequent chase water injection. This sequence was executed in the field as part of the LTPIT, during which surveillance data were gathered at each stage. These field measurements were then used to evaluate long-term polymer injectivity and to calibrate and history-match the simulation model. The calibrated sector model developed during the pilot phase was subsequently scaled up to represent the first-phase commercial development. This stage involved a multi-pattern injection system with irregular well spacing and closed-pattern polymer injection designs to optimize sweep efficiency. Finally, an economic analysis was performed to compare the benefits of polymer flooding relative to conventional waterflooding, confirming its viability for enhanced oil recovery.

Laboratory evaluation to select suitable polymer in harsh environment of reservoirs and effluent water as injection source

In EOR screening Kuwait's several sandstone and carbonate reservoirs identify potential for polymer flooding to accelerate oil production and increase oil recovery over water flood. KOC decided to cay out polymer injectivity tests in 6 sandstone reservoirs and one carbonate reservoir using produced/effluent water as an injection source. These reservoirs have harsh environments. However, in such extreme conditions, evaluating polymer stability becomes a critical factor. At high temperatures, chemical degradation can lead to polymer precipitation, particularly in the presence of divalent ions in the brine. Furthermore, thermal stability can significantly affect polymer viscosity under harsh conditions, especially when oxygen, iron, or polymer impurities are present. In conclusion, polymer stability is influenced by multiple factors, including —but not limited to—oxygen, salinity, hardness, temperature, polymer structure and molecular weight, polymer residuals, and ph. Therefore, an extensive laboratory evaluation carried out to select suitable synthetic polymer with different molecular weight and different mole% ATBS molecule from different vendors for harsh environment of these reservoirs as well utilize high TDS and hardness produced/effluent water as an injection source (Abdullah et al., 2023). In this methodology we will explain laboratory evaluation for case study 4 to select suitable polymer for this UGMO reservoir as follows:

Polymer/Brine Compatibility Testing (Polymer Stability). The polymer stability test visually evaluates the stability of the polymer under elevated temperature and salinity conditions of the reservoir. The polymer mother solutions at a reservoir temperature for 9 days to determine if the polymer is stable in high TDS saline water at high temperature. If visual observations confirmed that there was no formation of separate layers or settled down particles in the solution, then move further for another test as follows:

Filter Ratio Test. A polymer solution of target concentration was passed through A 47 mm diameter filter holder with 5 microns Millipore polycarbonate filter paper to measure variations in solution filterability due to undissolved solids and its proclivity to plug pores within the membrane. The test was done using 600 ml sample in room temperature and a pressure of 20 psi using nitrogen. The filtration results of different polymers are shown in plot (a) of Fig. 2

Screen Factor Test. The screen factor (SF) is defined as the ratio of the flow time of a polymer solution through a screen viscometer to the flow time of the same volume of solvent, typically water. It serves as an indicator of the viscoelastic behavior of the polymer, reflecting its ability to withstand sudden elongational deformation and the associated normal stresses. Screen factor measurements are directly related to the viscoelastic properties of the polymer solution. In this study, the test was conducted using a screen factor viscometer bulb, and the results are presented in Plot (b) of Fig. 2. Additionally, the screen factor can interpret as RF of the polymer, as shown below:

Screen Factor = RF = *Time taken for polymer solution/ Time taken for simulated water*

Apparent Viscosity Vs. Polymer Concentration. The polymer apparent viscosity versus polymer concentrations of 500, 1000, 1500, 2000, 3000 and 4000 ppm were tested using Brookfield LVDVNext viscometer, with a YULA-15EM spindle (UL Adapter) at a constant shear rate of 7.34 sec-1 (6 rpm) and a reservoir temperature controlled by a recirculating bath from PolyScience. The results are depicted in Plot (c) of Fig.2.

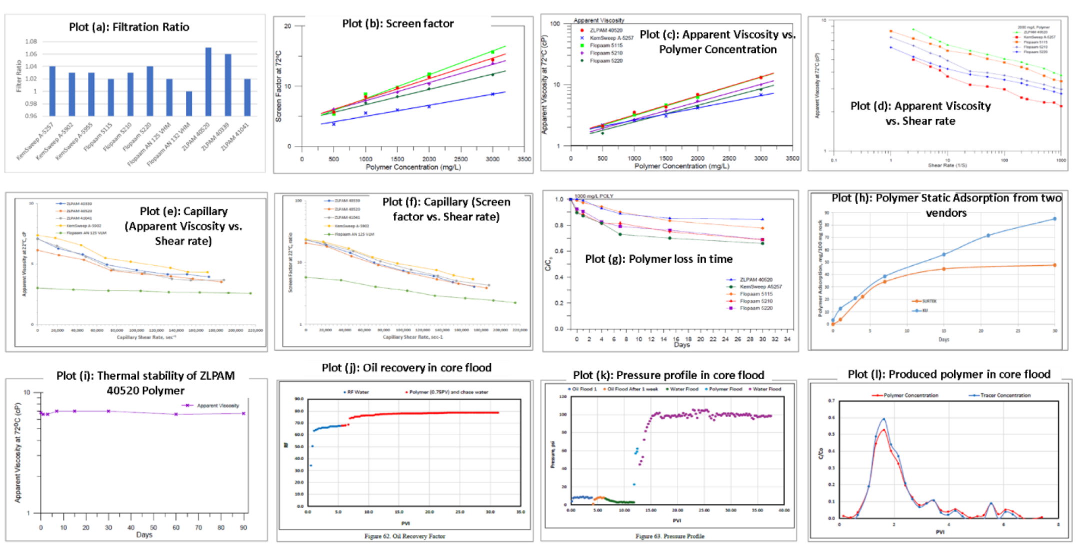

Figure 2—Results of different tests performed in laboratory

Apparent Viscosity Vs. Shear Rate. The polymer apparent viscosity with a range of shear rates up to 240 sec-1 was tested using Brookfield LVDVNext viscometer, with a YULA-15EM spindle (UL Adapter) concentration of 2000 ppm and a reservoir temperature controlled by a recirculating bath from PolyScience. The trends of apparent viscosity versus shear rate relationships show typical pseudoplastic behavior for all polymer solutions for the investigated shear rate range. From the experimental results of apparent viscosity versus shear rate tests, power-law model is evaluated according to the following equation: $\boldsymbol{\mu_{app}=K\gamma n\text{-}1}$

Where: Mapp, K, γ and n = Apparent viscosity (cp), flow consistency index (power law constant), shear rate (Sec-1) and power law exponent, respectively

If n = 1.0, then K remains constant at a fixed viscosity (μ), and the fluid behaves as a Newtonian fluid

If n ≠ 1.0, the shear stress does not vary linearly with the shear rate, and the fluid behaves as a non-Newtonian fluid.

Therefore, flow Consistency Index (K) correlates with the apparent viscosity, where higher K values represent higher apparent viscosity and vice versa. Table 2(a) and Plot (d) of Fig.2 shows the relationship of apparent viscosity with shear rate.

Table 2—K and N values of different polymer and hydration test results

Table (a): K &n value

Polymer Name	K	n
ZLPAM 41041	32.34	0.711
KemSweep A-5902	16.35	0.807
KemSweep A-5955	15.71	0.788
Flopaam AN 132 VHM	12.95	0.796
Flopaam AN 125 VHM	12.18	0.823
ZLPAM 40339	9.23	0.866
ZLPAM 40520	8.13	0.851
Flopaam 5115	8.03	0.881
Flopaam 5210	6.65	0.87
Flopaam 5220	6.45	0.858
KemSweep A-5257	4.38	0.931

Table (b): Dissolution/Hydration Test Results

Flopaam 5115	150
Flopaam 5210	150
Flopaam 5220	120
ZLPAAM 40520	120
KemSweep 5257	210
KemSweep A-5902	120
ZLPAM 40339	120
Flopaam AN 125 VHM	150
Flopaam AN 132 VHM	120
ZLPAM 41041	120
KemSweep A-5955	90

Table (c): Results of Thermal Stability Test

Chemical Solution	Days	Viscosity at 72 °C		Screen Factor at 72 °C	
		(cP)	% Change	Ratio	% Change
Flopaam 5115	0	7.2	0.0	27.3	0.0
	1	7.2	0.0	19.2	(29.8)
	3	7.1	(1.4)	18.1	(33.7)
	7	6.6	(8.3)	18.0	(34.2)
	15	7.2	0.0	18.4	(32.7)
	30	6.8	(5.6)	15.0	(45.1)
	60	5.7	(20.8)	16.5	(39.7)
	90	6.0	(16.7)	17.9	(34.5)
Flopaam 5210	0	6.0	0.0	19.3	0.0
	1	5.8	(3.3)	15.7	(18.5)
	3	5.7	(5.0)	14.3	(26.1)
	7	6.0	0.0	15.7	(18.6)
	15	6.3	5.0	16.3	(15.5)
	30	5.5	(8.3)	16.1	(16.8)
	60	5.0	(16.7)	17.8	(7.9)
	90	4.7	(21.7)	15.5	(20.0)
Flopaam 5220	0	5.1	0.0	15.2	0.0
	1	5.0	(2.0)	13.8	(9.7)
	3	5.0	(2.0)	13.6	(10.9)
	7	4.7	(7.8)	13.8	(9.7)
	15	4.8	(5.9)	14.2	(6.6)
	30	4.8	(5.9)	13.3	(12.3)
	60	5.0	(2.0)	14.2	(6.8)
	90	4.7	(7.8)	13.9	(8.6)
ZLPAM 40520	0	6.8	0.0	27.5	0.0
	1	6.6	(2.9)	24.2	(12.1)
	3	6.6	(2.9)	20.6	(25.0)
	7	7.0	2.9	18.9	(31.2)
	15	7.0	2.9	19.8	(28.1)
	30	7.0	2.9	16.4	(40.5)
	60	6.6	(2.9)	18.5	(32.9)
	90	6.7	(1.5)	17.5	(36.6)
KemSweep A-5257	0	4.3	0.0	18.8	0.0
	1	4.4	2.3	11.8	(37.4)
	3	4.3	0.0	11.0	(41.1)
	7	4.2	(2.3)	10.9	(41.7)
	15	4.8	11.6	10.9	(42.0)
	30	4.6	7.0	9.63	(48.7)
	60	4.3	0.0	12.3	(34.5)
	90	3.8	(11.6)	11.0	(41.1)

Dissolution/Hydration Test. The polymer dissolution/hydration test is designed to estimate the time required for a dry polymer to reach maximum viscosity, and hence complete hydration of the solid polymer. This was performed by monitoring apparent viscosity of the polymer solution over time. 5000 ppm polymer solutions were prepared at 21°C by mixing dry polymer with simulated water using a magnetic stirrer at 400 rpm, samples were removed at specified time intervals and viscosity measurements were recorded using a temperature-controlled viscometer, LVDVNext, at 2.64 and 6.60 sec-1 (2.2 and 5.4 rpm respectively) using YULA-15EM spindle (UL adapter). Then the sample was returned to the original mixture for continued mixing. This was repeated until apparent viscosity values asymptote. The time for the apparent viscosity to asymptote is a measure of polymer dissolution/hydration time. The result of this test is depicted in Table 2(b)

Capillary Shear Test. The stability of polymer solutions under reservoir conditions is crucial for polymer flooding success. Capillary shear test is conducted to evaluate polymer stability. Measurement of the effect of mechanical degradation on the polymer solution is important for determining the correct techniques for preparation, pumping and handling the solutions in the laboratory and in the field. The capillary shear test is one way to study the mechanical degradation in which a prepared polymer solution of concentration of 1000 ppm is pressurized through a capillary tube (0.0762 cm inner diameter) at different flow rates by using varying pressure levels. The objective of this test is to evaluate the effect of shearing on polymer solution stability through changes in viscosity. Shear rates range between 1000 and 100,000 sec-1 and can be calculated by recording fluid flow rates through the measured capillary. The apparent shear rate at capillary wall can be calculated as: $\gamma = 4Q/\pi R3 - Q = q/\rho$

Where: γ, q, Q, R and ρ: apparent shear rate at capillary wall (without non-Newtonian correction) in sec-1, flow rate (gm/sec), flow rate (cm3/sec), inside radius of capillary (cm) and density (gm/cm3), respectively

The polymer apparent viscosity measurements before and after every shearing step were recorded using a temperature-controlled viscometer, LVDVNext, using YULA-15EM spindle (UL adapter) at 22°C. The screen factor measurements were done at 22°C similarly. The results are shown in Plot (e & F) of Fig.2.

Static Adsorption Test. Static adsorption tests are conducted to examine polymer tendency to adsorbed to the rock surface at static conditions. This test provides an insight to compare different polymers about their potential retention in porous media. Static adsorption test can provide a preliminary screening of polymers and compare consumption of various polymers over time rather than for absolute values. This test is to estimate polymer losses that may occur due to adsorption of reservoir sand. The test was performed several times and presented on below subsections. The first test was performed on the crushed rock sample of grain size between 425 µm to 610 µm, received from core using 1000 ppm of polymer mother solution. A polymer solution with a known concentration is added to a sample of a crushed sand at 3:1 ratio by weight. The mixture of the polymer solution and the sand samples were kept on a mixing wheel and stored in oven at reservoir temperature. Samples of the polymer solution were taken at 0, 1, 3, 7, 15, 21 and 30 days' time interval and the concentration were measured using Starch Iodide method and measured polymer concentration over a different time interval results from Starch Iodide method. The results are shown in Plot (g & h) of Fig.2.

Thermal Stability. Five polymer solutions were evaluated for thermal stability at reservoir temperature when dissolved in Produced Water. Polymer concentrations are held constantly for all solutions. All chemical solutions were placed in ampules, oxygen was removed and replaced with argon by alternating vacuum and argon purge before the ampules were sealed with oxygen content of less than 50 ppb. All solutions were incubated in sealed ampules at reservoir temperature. Samples were removed from the oven on days 0, 1, 3, 7, 15, 30, 60, and 90. The results are shown in Table 2(c) and Plot (i) of Fig.2.

Core Flooding. Core flooding was carried out in different reservoir cores with an injection rate of 6 ft/day. Water was initially injected until no more water was produced from the core, which was then aged for one week. Oil flooding has been continued after the aging before shifting to water flooding at the same injection rate for more than 5 pore volumes. Corey models have been used to construct the oil-water relative permeability curve. After that, around 0.75 pore volume of target concentration polymer solution which contains 40 ppm of KI tracer have been injected with injection rate of 6 ft/day, followed by water flooding until no polymer was produced from the core. Polymer and tracer concentrations have been measured in the effluents using UV-Vis Spectrophotometer and the plot of normalized concentrations versus pore volume is shown in Plot (j) of Fig.2. Polymer retention was calculated using mass balance as 0.01 mg/g (3.58%). Plot(k), The core flood results for oil recovery factor and pressure profile constructed during this experiment are shown in Plot (k & l) of Fig.2.

Implementation of LTPIT in the field

During the execution of the LTPIT, the injection plan was developed based on outcomes from simulation-based pre-modelling. The implementation methodology and surveillance data interpretation for Case Study 1 (Al-Mayyan et al., 2025; 2023; 2022; 2019; 2018; 2017) will be discussed in detail. Initial model calibration was conducted using data collected during the baseline water injection period, as shown in Fig. 3. The LTPIT was carried out in three main stages: baseline water injection, polymer injection, and chase water injection. Over a period of 3 months, both water and polymer were injected into the well. To enable real-time pressure and temperature tracking, downhole sensors were deployed, and injection rates were continuously recorded. Prior to injection, feed water was treated to meet required standards (as listed in Table 3) and mixed with polymer using a compact dispersion system. Fig.3 also illustrates the typical injection steps across different phases. In the baseline phase, injection began at a low rate to stabilize BHP, followed by injection at the highest permissible rate, which was limited by estimated fracture pressure. A PFO test was then used to assess near-wellbore conditions, including skin factors and reservoir permeability. If high positive skin was detected, stimulation was advised to mitigate near-wellbore damage. SRT was also performed to determine the fracture initiation threshold. To assess the zonal distribution of injectivity, diagnostic tests including High-Precision Temperature (HPT), Spectral Noise Logging (SNL), and Injection Logging Tests (ILT) were performed. Polymer injections followed, beginning with a low concentration (e.g., 500 ppm), which gradually increased in four steps to reach the optimum value based on laboratory core flood analysis. Injection pressures and rates were carefully monitored to remain within matrix limits and below fracture pressure. At each polymer concentration stage, the rate was gradually increased to assess injectivity performance. Upon reaching the final concentration level, HPT- SNL-ILT was repeated to observe changes in the vertical injection profile relative to water-only injection. An additional SRT was performed, when feasible, to re-evaluate fracture pressure under polymer injection conditions. The final phase involved post-polymer chase water injection. A second PFO test was performed to re-measure reservoir permeability. By comparing permeability data from both the initial and post-polymer water injection phases, the residual resistance factor (RRF) was estimated to quantify the permeability reduction caused by polymer retention in the reservoir.

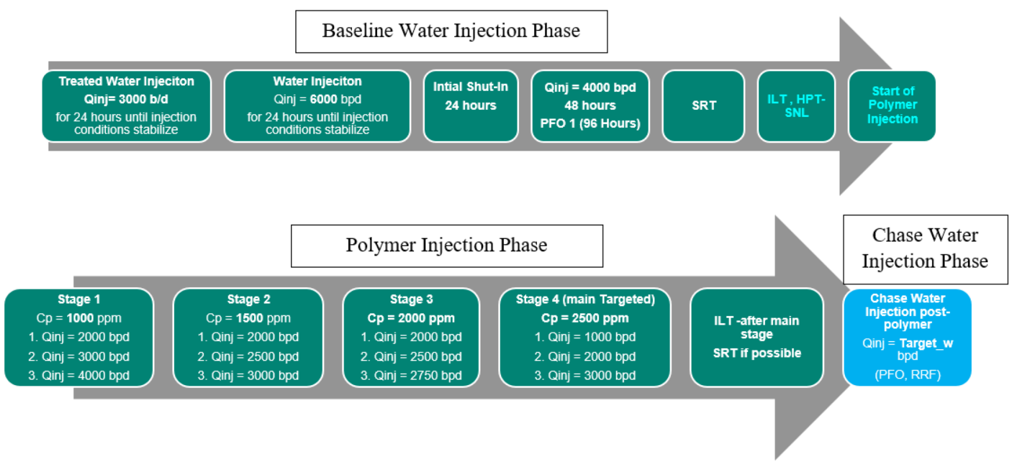

Figure 3—Injection sequences design based on pre-modeling

Table 3—Feed water targeted specs

Total Suspended Solids (TSS):	<5ppm
Oil in Water (OiW):	<10ppm
Dissolved Oxygen:	<10ppb
Iron content:	<1.5ppm
H2S:	<5ppm
Bacteria:	<10^3 counts/ml

Interpretation of collected surveillance data from LTPIT

Feed Water and Polymer Quality Monitoring. To ensure consistent performance during polymer injection, the quality of feed water was continuously monitored and analyzed for key parameters including Total Suspended Solids (TSS), Oil-in-Water (OiW), dissolved oxygen, iron concentration, hydrogen sulfide (H_2S), and microbial activity. These values were tracked in real time and maintained within the targeted specifications outlined in Table 3, as illustrated in Plots (a) to (d) of Fig.4. Simultaneously, the polymer solution quality was assessed by measuring the screen factor, filtration ratio, and viscosity, shown in Plot (d) of the same figure. These metrics ensured the polymer maintained consistent rheological properties and injectivity performance.

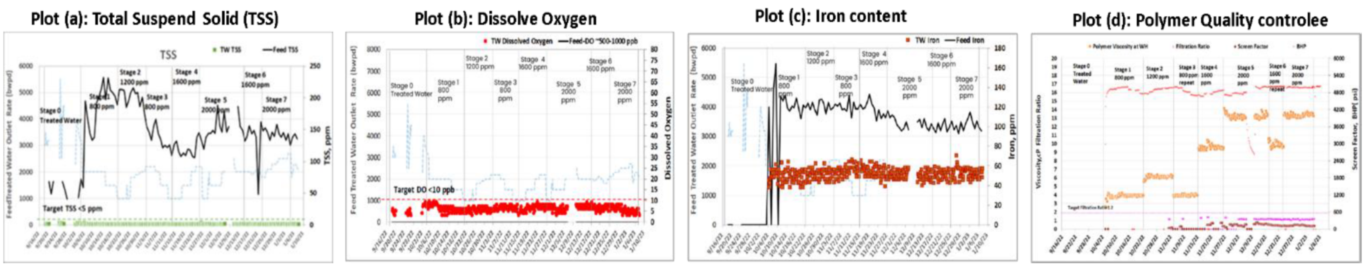

Figure 4—Injected feed water and polymer quality control to meet target specs

Monitoring Polymer Breakthrough

Produced fluid samples from wells near the LTPIT site were periodically collected to detect any polymer breakthrough. No evidence of polymer presence was detected in any of the surrounding producers, indicating strong conformance control. The leading edge of the polymer front was observed to reach approximately 200 meters from the injector in the best injectivity scenario at 1000 ppm concentration. A structured injection plan was implemented, involving stepwise increases in polymer concentration and flow rates to evaluate injectivity at each stage. This gradual ramp-up helped assess polymer propagation and behavior within the reservoir.

Baseline Water Injection, RF, and RRF Evaluation

During the baseline phase, various surveillance tools—including SRT, PFO and HPT-SNL-ILT were used to collect reservoir response data. SRT results, as shown in Plot (b) of Fig.5, were analyzed to establish the fracture pressure threshold, which was then used to define safe injection pressures. PFO data from this phase were interpreted to estimate reservoir permeability and skin factor (Plot b, Fig.5). When high skin was observed, acid stimulation was performed to improve near-wellbore conditions. Table 4 outlines the definitions, equations, and significance of RF and RRF in evaluating polymer flood performance. RF, tracked over time, reflects changes in flow resistance due to polymer presence, while RRF quantifies permeability reduction resulting from polymer retention. If RF>1throughout the pilot period, confirming that no fracture-induced flow paths were created. RRF was determined by comparing water permeability before and after polymer injection interpreted from PFO tests conducted during water injection pre and post polymers injection as seen in Plots (b) and (c) of Fig.5. The transition from a polymer-swept zone to a water-dominated region was also observed with increasing radial distance from the wellbore. A notable reduction in permeability from 29.5 mD to 12.2 mD corresponded to an RRF of approximately 2.4. This value, calculated over the radius of investigation, aligns with expectations and differs from the sand-face RRF of 1.5 discussed earlier.

Table 4—Definitions and description of RF and RRF

Resistance Factor (RF)	Residual Resistance Factor (RRF)
Darcy Law: $q = \frac{KA}{\mu}\frac{\Delta p}{L} \Rightarrow K = \frac{\mu}{A}\frac{q}{\Delta p}$; Mobility $(\Lambda) = \frac{K}{\mu}$	Mobility Ratio (MR) $= \frac{\Lambda_i}{\Lambda_d}$
$RF = \frac{(\Delta p/q)_{p\,(polymer\ injection\ phase)}}{(\Delta p/q)_{w\,(base\ line\ water\ injection\ phase)}} = \frac{\Lambda_w}{\Lambda_p}$	$RRF = \frac{(\Delta p/q)_{w\,(Chase\ water\ injection\ phase\ after\ polymer)}}{(\Delta p/q)_{w\,(base\ line\ water\ injection\ phase)}} = \frac{\Lambda_{w\,(Chase\ water)}}{\Lambda_{w\,(base\ line\ water)}} = \frac{K_{w\,(chase\ water)}}{K_{w\,(base\ line\ water)}}$
▪ Injectivity loss and plugging of the reservoir matrix ▪ Mobility Reduction ▪ Example: If RF=3 then injectivity loss by well will be by factor 3, hence injection pressure in polymer phase increase 3 times to maintain same injection rate as water phase or polymer injection rate will be reduces as $\frac{1}{3}$ compare to water phase to maintain same pressure in both phases	▪ Polymer retention by monitoring breakthrough in nearby producer(s) ▪ It evaluate polymer effectiveness ▪ It's a ratio mobility of water phase before and after polymer flood ▪ It is also representing permeability reduction in chase water phase compared to base line water phase ▪ Permeability reduction also will be compared from PFO in chase water and base line water.

Note: Pressure and rate should be measure in steady state condition i.e. in water flood phase at S_{orw} so that saturation will not change after that by water flood

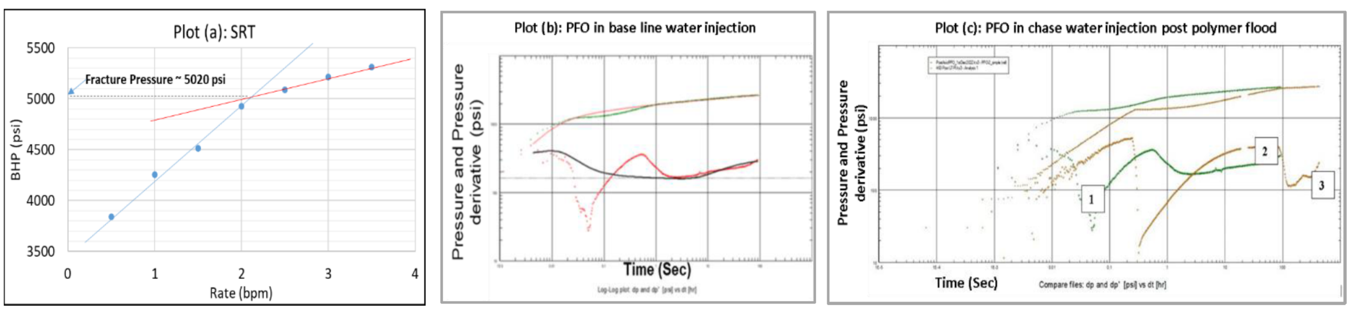

Figure 5—SRT and PFO performed in base line and chase water injection phases

Vertical Injection Profiling (HPT-SNL-ILT)

The HPT-SNL-ILT diagnostic suite was instrumental in assessing zonal flow distribution before and after polymer injection. Post-injection results, shown in Fig.6, indicated a marked improvement in vertical conformance. Initially uneven fluid intake across reservoir zones became more uniform following polymer treatment, highlighting the effectiveness of the intervention in enhancing sweep efficiency and vertical coverage.

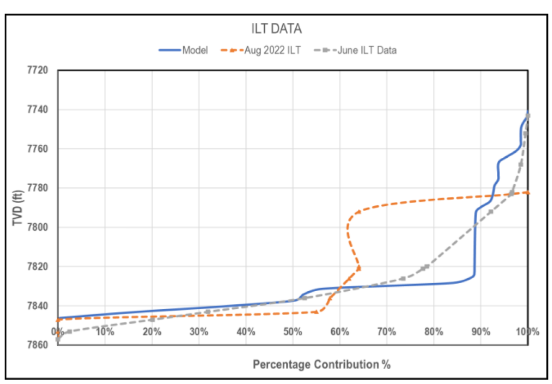

Figure 6—Downhole injection profile comparison pre (June) and post (Aug) polymer injection

Injectivity Performance and Estimated Flowing Viscosity of Polymer Solutions

Figure 11, Plot (s), illustrates the trends in injection pressure, injection rate and polymer concentration over the entire pilot duration. Polymer injections began at a conservative concentration of 800 ppm to mitigate the risk of formation plugging. Subsequently, concentrations increased to 1200, 1600, and 2000 ppm. At each concentration, the multiple injection rates were tested. Each stage was maintained until the system reached a steady-state pressure or exhibited an asymptotic trend. The near wellbore flowing viscosity of polymer solutions was estimated using the RRF, which inherently accounts for changes in effective water permeability. This methodology allows comparison of the effective in-situ viscosity with lab-measured rheometer values. A base water viscosity of 1.2 cP was used for viscosity calculations. The effectiveness of the LTPIT was therefore evaluated by correlating the injection rates, apparent downhole viscosities at various polymer concentrations, and the deviation from laboratory results.

Flowing Viscosity versus Injection Rate. Estimated flowing viscosities were observed to be higher at lower injection rates, corresponding to lower shear rates, which is characteristic of pseudoplastic or shear-thinning behavior of the fluids. These trends became more prominent when viscosities were plotted against calculated shear rates, using a cylindrical model of the perforations and assuming a 32 ft effective injection interval based on HPT-SNL-ILT diagnostics. Plot (a) of Fig.7 demonstrates the viscosity variations during different polymer injection stages, along with baseline and chase water phases. The viscosity data confirmed that polymer solutions were flowing through the reservoir matrix rather than through induced fractures. This was evidenced by higher viscosities at lower injection rates, which would not occur if fractures were present.

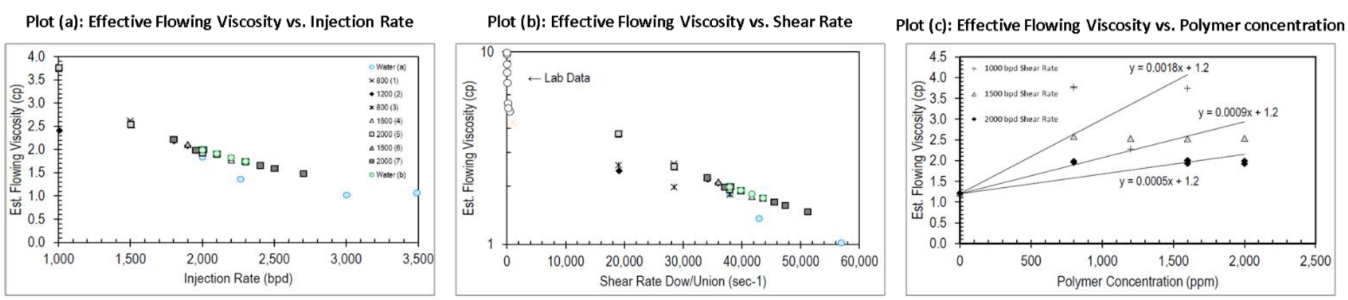

Figure 7—Effective flowing viscosity relationship with injection rate, shear rate and polymer concentration

Apparent Downhole Viscosity Trends. At higher shear rates, the estimated viscosities for various concentrations converged toward water-like values, indicating shear thinning in the near-wellbore region. This behavior further supports the absence of mechanical degradation of the polymer, as damaged polymer would result in viscosity values remaining low across all shear rates. The estimated viscosities for 800 ppm injection matched well with the laboratory data, and although the calculated field values had slightly steeper slopes due to the near-wellbore effects, they remained consistent with lab-measured rheology. These in-situ viscosity estimates, when plotted against polymer concentration for varying injection rates (e.g., 1000, 1500, and 2000 bbl/d), showed an increase in apparent viscosity with polymer concentration, as shown in Figure 7, Plot (c). However, under high shear conditions (associated with higher injection rates), this trend became non-linear, indicating a limit in the observable viscosity changes due to shear effects. Lower injection rates provided more pronounced viscosity increases, especially in the Upper Burgan reservoir. The 800 ppm pre-stimulation data exhibited expected trends: higher injection rates led to reduced flowing viscosities due to elevated shear. The post-stimulation data followed similar trends, though the data quality declined due to low bottomhole pressures (vacuum conditions). For instance, the 1600 ppm stage was excluded from the analysis because the pressure exceeded fracture limits, potentially affecting polymer performance. In comparing the 800-ppm case at different rates, the 1000 bpd scenario produced a higher viscosity than at 1500 bpd, affirming non-Newtonian flow characteristics. Lab-measured data for 800 ppm aligned well with field estimates, confirming that the field conditions preserved polymer properties during injection. Figure

7, Plot (a), further compares the calculated shear rate-based viscosities using the Dow/Union correlation, with field measurements closely matching lab results. This indicates that shear-thinning behavior observed in the lab is reflected under field injection conditions.

Confirmation of Pseudoplastic Behavior. The rheological analysis near the injection well confirmed that the polymer exhibited pseudoplastic behavior, with increased in-situ viscosity at higher concentrations. Field-observed viscosities ranging from 4.6 to 4.8 cP at 1600 ppm were comparable with the lab-designed value of 3.3 cP, validating the consistency and performance of the polymer solution under reservoir conditions.

Simulation modeling for first phase commercial polymer flooding

An extended area surrounding the LTPIT site was assessed for potential expansion of polymer flooding. However, due to the absence of reliable injectivity data, significant heterogeneity in permeability-thickness (KH), and unresolved geological uncertainties, this area was deprioritized for full-field development (Fig.8). Therefore, based on current reservoir condition, high opportunity index area lies in northwest to the crest of the reservoir on the flank approaching the aquifer. Most of this area lies between two major faults. Also, this pilot area shows higher current remaining oil saturation, higher oil thickness and good pay-zone thickness. Opportunity index defined as:

$$OI_{sum} = \sum_{L=27}^{L=38} \left[H * \left(1 - S_{w,end\ HM}\right) * log(k) \right]$$

Where H, $S_{w,end\ HM}$ and k are the reservoir thickness, current water saturation and permeability, respectively.

A revised sector model for the new Area of Interest (AOI) was developed and history-matched using updated injection and production data. The model was further calibrated using recent Production Logging Tools (PLTs) and Injection Logging Tests (ILTs) to accurately replicate flow contributions from various Upper Burgan (UB) sublayers. Based on this calibration, the AOI was identified as suitable for phased commercial-scale polymer flood development. Multiple development configurations were simulated to maximize incremental oil recovery, with a focus on leveraging existing infrastructure, wells and facilities to minimize capital costs. Irregular pattern development emerged as the optimal strategy, allowing the integration of existing wells while avoiding extensive drilling or abandonment. Irregular injection pattern-based development was considered less practical due to its requirement for significant new drilling and potential conflicts with historical well placements. Therefore, irregular spot injection pattern was formulated utilizing maximum existing wells with typical 200-400 well spacing for polymer flooding. Initially, six irregular injection patterns were designed for detailed evaluation. Further analysis, including production forecasts and economic screening, led to the selection of five optimal patterns for in-depth modeling and cost-benefit assessment. This plan included three infill injector wells and one new producer well, supplementing two additional wells that were scheduled for commissioning within the following six months. The final five-pattern layout incorporated 13 existing wells and five new wells (four injectors and one producer).

Simulation results indicated that the optimized five-pattern configuration could achieve a cumulative incremental oil recovery of 7.3 %OOIP over the conventional water flood based on 0.3 PVI polymer injection, which corresponds to an incremental cumulative oil of 2.8MMST. Economic analysis incorporated key components, including injection water supply, produced water treatment, and new well drilling. The Unitary Technical Cost (UTC) was estimated at $17.32/bbl over the waterflood case, affirming the economic viability of polymer implementation. Early deployment of polymer, alongside infill drilling, was found to enhance oil recovery by improving areal sweep efficiency and reducing excessive water production.

Waterflooding simulations revealed unfavorable mobility ratios (greater than 1) across all patterns, especially Patterns 2 and 3, due to low oil mobility and high-water mobility. In contrast, polymer injection significantly improved displacement efficiency, reducing mobility ratios closer to unity in most patterns, except for Pattern 2, which remained marginal. These findings reaffirmed the non-Newtonian, pseudoplastic behavior of the polymer, enhancing sweep efficiency in matrix flow conditions without causing formation damage.

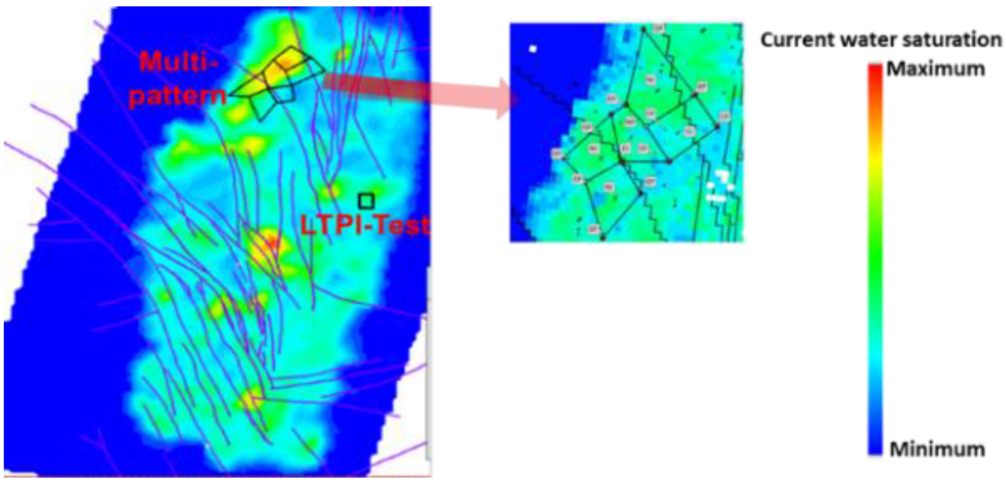

Figure 8—Location of LTPIT and first phase commercial polymer flooding

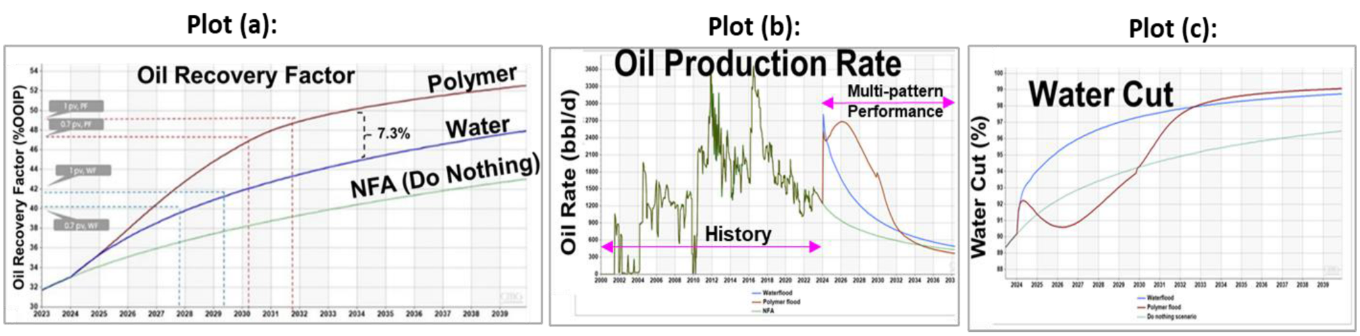

Figure 9—Polymer flooding performance of first phase commercial polymer flooding

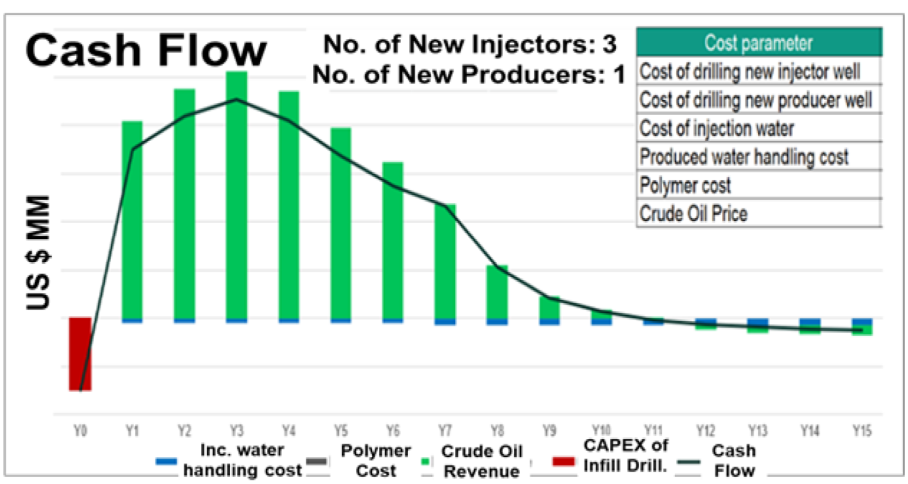

Figure 10—Economic Analysis

Results and Discussions

Challenges, and overcome and results of implemented LTPIT in the field from different case studies

The produced water, characterized by high salinity, was treated to maintain dissolved oxygen levels below 10 ppb, ensuring that the iron content did not adversely affect the polymer's viscosity. Therefore, the feed water target specs as specified in Table 3 were maintained before injection of water or polymer in all LTPIT of different case studies as illustrated in Table 5. Then LTPIT will perform in three phases as base line water, polymer and chase water injection phases. The results of different LTPIT will be discussed as follows:

Table 5—Mobility ratio calculation for polymer flooding

Pattern	Oil Rel Perm	Oil Viscosity, cp	Water rel perm	water viscosity, cp	Poymer viscosity, cp	Mobility Ratio in Water Flood	Mobility Ratio in Polymer Flood
1	0.09480	3.90430	0.07220	0.68130	4.6	**4.4**	**0.6**
2	0.05653	4.96000	0.23199	0.68388	4.6	**29.8**	**4.4**
3	0.08930	3.62892	0.11092	0.67870	4.6	**6.6**	**1.0**
4	0.08222	4.13790	0.12730	0.68046	4.6	**9.4**	**1.4**
5	0.14559	3.69851	0.05929	0.67126	4.6	**2.2**	**0.3**
Total	0.09345	3.91420	0.11902	0.67910	4.6	**7.3**	**1.1**

North Kuwait (Case study 1, 2 & 3).

Case study 1

The LTPIT was carried out in SAUB reservoir (Al-Mayyan et al., 2025; 2023; 2022; 2019; 2018; 2017). This LTPIT pilot demonstrated effective polymer injectivity over a period of 3.6 months, with injection rates of 1,000, 1,500, and 2,000 bbl/d as shown in Plot (a) of Fig.11. These injections were conducted using produced water with polymer concentrations of 800, 1000, 1500 and 2000 ppm. Near the wellbore region, the polymer solution exhibited pseudoplastic behavior under matrix flow conditions, as the in-situ viscosity increased with higher polymer concentrations. At a concentration of 1600 ppm, the polymer achieved an in-situ viscosity of approximately 4.6–4.8 cP, which aligns closely with the laboratory-designed target of 3.3 cP. Pressure fall-off (PFO) test results before and after polymer injection indicated a reduction in reservoir permeability from 29.5 mD to 12.2 mD, corresponding to a RRF of 2.4 within the radius of investigation, which is slightly higher than the sand-face RRF (1.5).

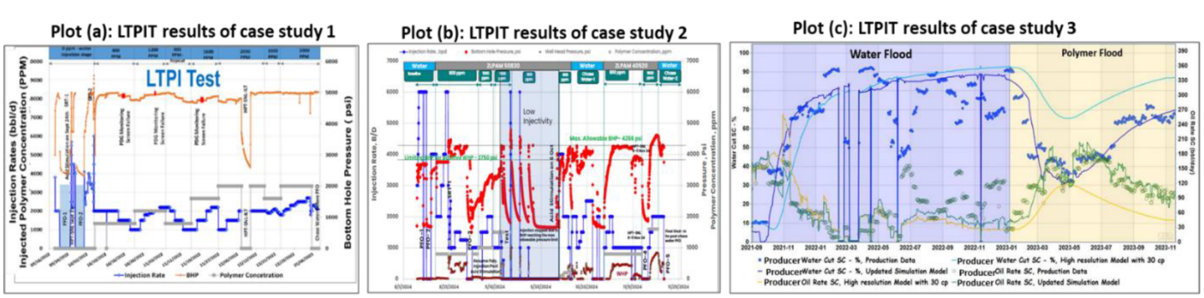

Figure 11—Results of implemented LTPIT from case studies 1, 2 & 3 in North Kuwait reservoirs

Case study 2

The LTPIT was carried out in SAUB reservoir (Abduraheem et al., 2025). This LTPIT pilot encountered injectivity issues as shown in Plot (b) of Fig.11, with the initially selected polymer (ZLPAM 50830) primarily due to its relatively high molecular weight (10–15 mD) and high ATBS content (50 mol%). The injectivity tests were conducted over 3.6 months with varying injection rates of 800, 1000, 1500, and 1800 barrels per day. When 3000 bbl/d polymer solution was injected at a concentration of 1200 ppm, then injection rate was gradually reduced to 2000 bbl/d and then to 1000 bbl/d due to reaching the injection pressure (fracture pressure) limit of 4266 psi. Subsequently, SRT and PFO tests were conducted to determine fracture pressure, reservoir permeability and near wellbore skin. Injection was resumed at 1000 bbl/d with 1200 ppm polymer, increasing incrementally by 250 bbl/d in every three hours, eventually reaching 2500 bbl/d. Acid stimulation was also performed to remove near-wellbore damage: however, injectivity issues with ZLPAM 50830 persisted. As a result, the decision was made to switch to ZLPAM 40520 as it has lower molecular weight (8–10 mD) and a reduced ATBS content (15–20 mol%). Then this polymer started injecting concentration of 800 ppm, followed by 1600 ppm at a flow rate of 1500 BPD. Key learnings from this pilot include:

- Bottomhole pressure (BHP) is significantly impacted by acid stimulation treatments.
- Downhole viscosity trends at 800 ppm concentration confirmed the expected non-Newtonian, pseudoplastic behavior of both polymers.
- The viscosity profiles also suggest no evidence of fracture flow at this concentration.
- Estimated shear rates matched well with laboratory-measured shear behavior, confirming that downhole viscosities can be extrapolated reliably from rheometer data.
- ZLPAM 50830 showed slightly higher resistance factors than ZLPAM 40520, consistent with its higher molecular weight.

Case study 3

The LTPIT was carried out in UNLF heavy oil reservoir (Al-Mayyan et al., 2024; 2018). In this study, a polymer flooding pilot was conducted using a regular five-spot injection pattern in a heavy oil reservoir as shown in Plot (c) of Fig. 11. The reservoir demonstrated excellent polymer injectivity and yielded highly encouraging results from the polymer flood. The selected polymer ZLPAM 40339 provides desirable in-situ viscosity as 11cp at economic polymer concentration of 1600ppm. During the initial waterflooding phase, a recovery factor of approximately 25% of the original oil in place (OOIP) was achieved after injecting 0.25 pore volumes of water, confirming the viability of using sour water for injections. Following the polymer injection phase, the oil cut increased significantly from 10% to 70%, resulting in an estimated incremental oil recovery of approximately 31% OOIP with just 0.31 PV polymer injection. The pilot also confirmed that the maximum allowable injection pressure could be safely increased from 60 psi to 120 psi without compromising the integrity of the cap rock.

West Kuwait (Case study 4 & 5).

Case study 4

The LTPIT was carried out in UGMO carbonate reservoir (Al-Mayyan et al., 2024; 2019; Eadulapally et al., 2024; Khan et al., 2023; 2024; Yunus et al., 2023). This LTPIT pilot demonstrated excellent polymer injectivity over a 6 month period, with injection rates ranging from 1,000 to 6,000 bbl/d as shown in Plot (a) of Fig. 12. Produced water was used as the injection medium, with polymer concentrations of 1000, 1500, 2000 and 2500 ppm. During the base line water injection phase, the well remained under vacuum conditions, transitioning gradually into the fill-up stage, which was marked by a corresponding rise in BHP. Overall, BHP trends showed a clear increase with both injection rate and polymer concentration. For each polymer concentration, stable BHP profiles were observed as injection rates were incrementally increased. In the near-wellbore region, the polymer solution displayed typical pseudoplastic behavior under matrix flow conditions, with in-situ viscosity increasing proportionally to concentration. At 2500 ppm concentration of polymer, the estimated in-situ viscosity reached approximately 7.5 cP. Across all concentrations, injectivity showed gradual improvement over time, largely associated with the ramp-up in injection rates. BHP exhibited a strong correlation with injection rate, generally rising initially and then stabilizing when both rate and concentration were held constant. RF calculated as the ratio of ($\Delta P/Q$) for polymer-injected fluid to that for baseline water were tracked over time. During the fill-up stage, prior to polymer injection, RF remained nearly constant. With the onset of polymer injections, the RF became responsive to changes in both injection rate and polymer concentration. As expected, injectivity declined by increasing polymer molecular weight and concentration. Throughout the pilot, the RF remained above 1, indicating effective polymer mobility control, without any sudden or unexpected injectivity surges. Importantly, injection BHP consistently remained below the fracture pressure, indicating that no fracture initiation occurred during the pilot period.

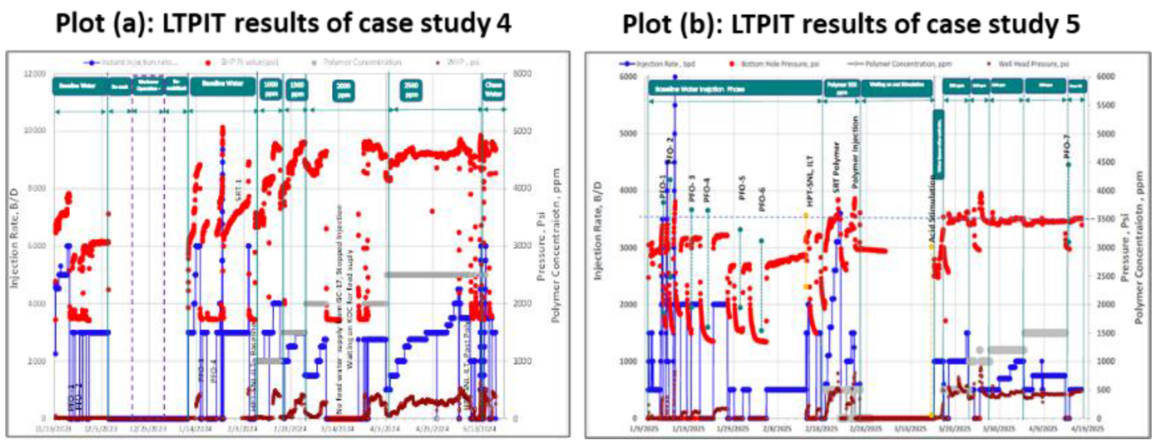

Figure 12—Results of implemented LTPIT from case studies 4 & 5 in West Kuwait reservoirs

Case study 5

This LTPIT pilot demonstrated effective polymer injectivity over a period of 3.6 months, with injection rates ranging from 800 to 3,500 bbl/d as shown in Plot (b) of Fig. 12. Produced water was used as an injection fluid, with polymer concentrations of 500, 1000, 1200 and 1500 ppm. In the near-wellbore region, the polymer solution exhibited pseudoplastic behavior under matrix flow conditions, where the estimated in-situ viscosity increased as the injection rate or shear rate decreased. At a concentration of 1500 ppm, the in-situ viscosity was estimated to be 4.8 cP, which is close to the laboratory-designed target viscosity of 5 cP. PFO tests conducted before and after polymer injection indicated a reduction in effective permeability, suggesting a RRF within the radius of investigation. However, signs of fracturing were observed even at relatively low BHP (~3200 psi), introducing some uncertainty in the interpretation. For each injection rate or shear condition, a consistent trend across concentrations was observed, which may be attributed to:

- **Flow through fractures**: Some PFO results showed evidence of fractures even at lower BHPs.
- **Improved reservoir connectivity**: Model-based analysis and PFO responses suggest the possibility of the reservoir connecting to a higher-permeability zone located 300–350 feet away, potentially activated during prolonged polymer injection.

As a result, post-polymer PFO tests exhibited a more pronounced linear flow regime, indicative of induced fracture effects. A drop or discontinuity in the pressure derivative around the 3-hour mark was observed, likely to correspond to a transition from fracture-dominated flow to reservoir flow. It is possible that the limited volume of chase water (~1000 barrels) was insufficient to fully displace the polymer slug, and the presence of fractures may have masked any clear reservoir response from the chase water. Therefore, due to the limited duration of chasing water injection, a clear radial flow regime was not observed in the PFO. Therefore, the RRF during the polymer injection phase was analyzed over time and consistently remained above 1, corresponding to higher bottom-hole pressure compared to the baseline water injection. The baseline pressure-to-flow (P/Q) ratio was established during stable water injection at 2000 bbl/d with positive WHP. Subsequent lower-rate injections showed significantly higher resistance, sometimes with negative WHP. Interestingly, a lower RF was observed at 500 ppm concentration injected at 1000 bbl/d (post-acid stimulation) compared to a similar water injection phase at ~500 bbl/d. This may indicate the use of near-wellbore microfractures for flow or reflect the effects of acid stimulation. As anticipated, growing the polymer concentration to 1000 ppm led to a higher RF. However, with prolonged polymer injections, the RF gradually declined at each concentration level, indicating possible polymer retention over time.

South and East Kuwait (Case study 6, 7, 8 & 9).

Case study 6

The LTPIT was carried out in Eastern Ware Sand of the Greater Burgan field (AlAbdullah et al., 2025; Al-Mayyan et al., 2022; 2023; 2024). This LTPIT pilot demonstrated effective polymer injectivity over a period of 3.6 months, with injection rates of 2,000 to 4500 barrels per day as shown in Plot (a) of Fig.13. These injections were conducted using produced water with polymer concentrations of 800, 1000, 1200, 1500, 1800 and 2000 ppm. Near the wellbore region, the polymer solution exhibited pseudoplastic behavior under matrix flow conditions, as the in-situ viscosity increased with higher polymer concentrations. At a concentration of 1500 ppm, the polymer achieved an in-situ viscosity of approximately 5 cP, which aligns closely with the laboratory-designed target. Pressure fall-off (PFO) test results before and after polymer injection indicated a reduction in reservoir permeability estimate RRF of 1.17 within the radius of investigation.

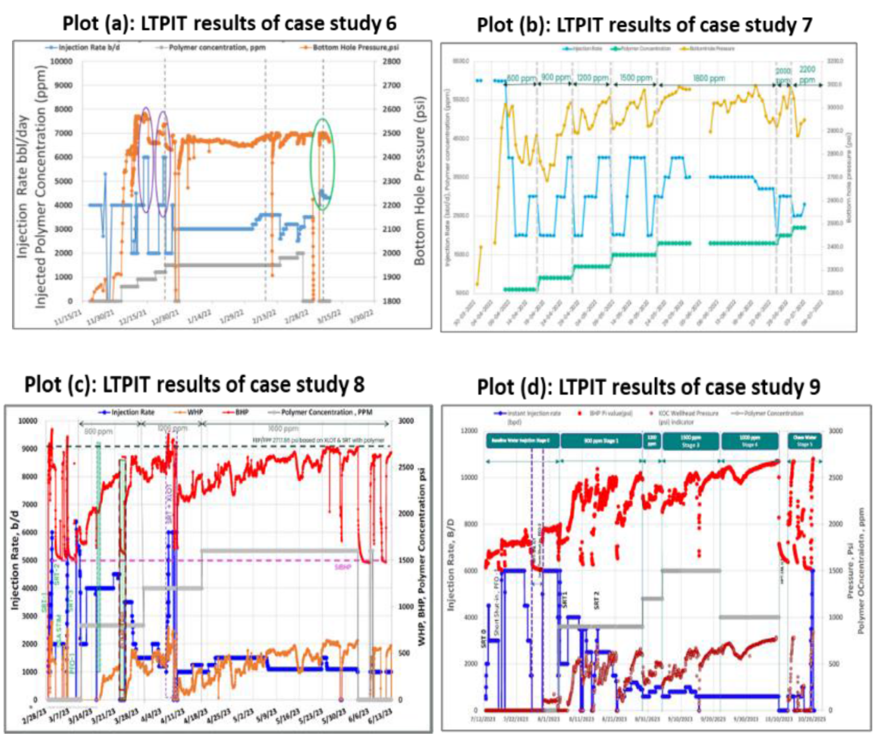

Figure 13—Results of implemented LTPIT from case studies 6, 7, 8 & 9 in South and East Kuwait reservoirs

Case study 7

The LTPIT was carried out in Western Ware Sand of the Greater Burgan field (AlAbdullah et al., 2025). This LTPIT pilot demonstrated effective polymer injectivity over a period of 3.6 months, with injection rates of 1000, 2,000, 2500, 3000, 3500 and 4000 barrels per day as shown in Plot (a) of Fig.13. These injections were conducted using produced water with polymer concentrations of 500, 1000, 1200, 1500, 1800, 2000 and

2200 ppm. Near the wellbore region, the polymer solution exhibited pseudoplastic behavior under matrix flow conditions, as the in-situ viscosity increased with higher polymer concentrations. At a concentration of 1800 ppm, the polymer achieved an in-situ viscosity of approximately 7 cP, which aligns closely with the laboratory-designed target. The PFO test results before and after polymer injection indicated a reduction in reservoir permeability estimate RRF of 2 within the radius of investigation.

Case study 8

The LTPIT was carried out in MG reservoir of Northern Ware Sand of the Greater Burgan field (Al-Qattan et al., 2024). The LTPIT pilot experienced injectivity challenges, as indicated in Plot (c) of Fig.13. These issues may have been triggered by events that negatively impacted injection performance. Polymer injections were conducted over 3.6 months at flow rates of 1,000, 1,500, and 2,000 barrels per day using treated produced water with polymer concentrations of 800, 1200, and 1600 ppm. Injectivity performance was comparable between 1200 and 1600 ppm until May 5. After changing to 1600 ppm, a gradual decline in RF was noted, followed by a sharp increase in both RF and BHP despite reducing the injection rate to 1100 bpd. The combination of 1600 ppm at 1500 bpd yielded the most favorable injection performance. Although variations in BHP were monitored alongside changes in injection rate and concentration, no consistent correlation emerged. Over time, BHP generally increased and then stabilized when injection parameters were held constant. However, interventions such as rate reductions when BHP reached its upper threshold disrupted this trend. To further understand flow behavior, resistance factors were calculated using the pressure-per-rate ratio ($\Delta P/Q$) for injected fluids compared to baseline water. These baseline values were taken from pre-acid stimulation conditions, introducing some uncertainty in the base RF estimate. The well-likely sustained damage during the campaign, as it became increasingly sensitive to operational changes. Despite elevated BHPs, resistance factors remained above 1, suggesting matrix flow without evidence of fracture propagation. Concerns arose about possible fracture reactivation, especially when BHP exceeded 2700 psi and erratic pressure behavior followed. An SRT-XLOT test was performed to investigate fracture pressure, but low-resolution data limited the ability to quantify fracture behavior. Post-injection PFO data did not indicate induced fracturing. As expected, apparent viscosities were higher at lower injection rates due to lower shear rates. However, at high shear rates near the wellbore, viscosity differences across polymer concentrations were minimal and close to that of water, indicating the polymer solution was not degraded mechanically during injection. The solution exhibited pseudoplastic behavior under matrix flow conditions, with increasing viscosity at lower shear rates across all concentrations. Laboratory rheology data confirmed effective flowing viscosity trends. On the surface, a concentrated polymer solution was initially prepared and then diluted using treated water to reach the desired injection viscosity. With no shear influence on the surface, higher polymer concentration consistently results in higher viscosities. In the reservoir, particularly near the wellbore, the polymer solution maintained its pseudoplastic nature, reaching an in-situ viscosity of about 3.85 cP at 1200 ppm—below the lab-designed target. PFO tests conducted before and after injection suggested a reduction in reservoir permeability, with a RRF estimated between 1.8 and 2.2 within the investigated zone.

Case study 9

This LTPIT pilot demonstrated effective polymer injectivity over a period of 3.6 months, with injection rates of 1000 6000 barrels per day as shown in Plot (a) of Fig.13. These injections were conducted using produced water with polymer concentrations of 1000, 1200 and 1500 ppm. Near the wellbore region, the polymer solution exhibited pseudoplastic behavior under matrix flow conditions, as the in-situ viscosity increased with higher polymer concentrations. At a concentration of 1200 ppm, the polymer achieved an in-situ viscosity of approximately 3.6 cP, which aligns closely with the laboratory-designed target. The PFO test results before and after polymer injection indicated a reduction in reservoir permeability estimate RRF of 3.5 within the radius of investigation, which acceptable in the field.

The findings from various LTPIT, as summarized in Table 6, indicate that most case studies demonstrated effective long-term injectivity of polymers under matrix flow conditions. The tested polymers exhibited pseudo-plastic rheological behavior, where an increase in polymer concentration (i.e., consequently in-situ viscosity) led to a corresponding rise in downhole pressure. Following extensive laboratory evaluations and core flooding experiments, the most suitable synthetic polymers were identified to contain 15–50 mole% ATBS molecule and 8-18 mD molecular weights, tailored to specific reservoir conditions and permeability levels. These selected polymers showed relatively low dynamic retention values, typically between 10 and 65µg/g of rock. Furthermore, the chosen polymer concentrations (1200 - 2500 ppm) were considered economically feasible, providing in-situ target viscosities 3.3 -7 cP. This viscosity range is sufficient to achieve a favorable mobility ratio compared to conventional waterflooding. Across all case studies, RRF remained within acceptable limits, ranging from 1.1 to 3.5.

Table 6—Quality control to maintain targeted specs for high TDS injected water in different case studies

Parameter	Case Study 1	Case Study 2	Case Study 3	Case Study 4	Case Study 5	Case Study 6	Case Study 7	Case Study 8	Case Study 9
TDS (ppm)	134,000	140,000	123867	190,000	261,200	210,000	170,000	127,600	89,000
Iron (ppm)	222	75	75	75	81	7.3	6.9	25	15
OiW (ppm)	110	5	5	8.5	6	10	6	10	31
Bacteria (cfu/ml)	15400	150	150	100	170	<1.8 mpn/100ml	10	3000	400
TSS (ppm)	294	1080	1080	1336	1190	160	142	268	139
H2S (ppm)	0.055	0.2	0.2	32	0.5	2.8	2	0.0078	0.007
Dissolve oxygen (ppb)	>5000	>5000	>5000	>5000	>5000	8000	>5000	8000	5800

Results of first phase commercial polymer flooding from different case studies

Following the successful completion of the LTPIT, and the subsequent calibration and history matching of the sector simulation model using surveillance data, a suitable area was identified for the first phase commercial polymer flooding. The selection was based on key reservoir characteristics, including high remaining oil saturation, good inter-well connectivity, favorable permeability, substantial net pay thickness and a high current opportunity index. The selected area was carefully delineated to maximize the use of existing wells, enabling the implementation of an irregular injection pattern, which contributes to better economic performance by reducing additional well drilling requirements. Regions with significant faulting, natural fractures, or strong aquifer support were intentionally excluded due to the potential risks they pose to sweep efficiency and polymer retention.

The selected areas for the first phase of commercial polymer flooding are illustrated in Fig.14 to 16, covering various reservoirs located in North, West, South, and East Kuwait. Fig.17 to 19 presents the relationship between oil recovery and pore volume injected (PVI) for both waterflood and polymer flood scenarios from different case studies. These figures clearly demonstrate that polymer flooding consistently results in accelerated oil recovery compared to conventional waterflooding. However, the magnitude of incremental oil recovery varies depending on the specific reservoir characteristics in each case.

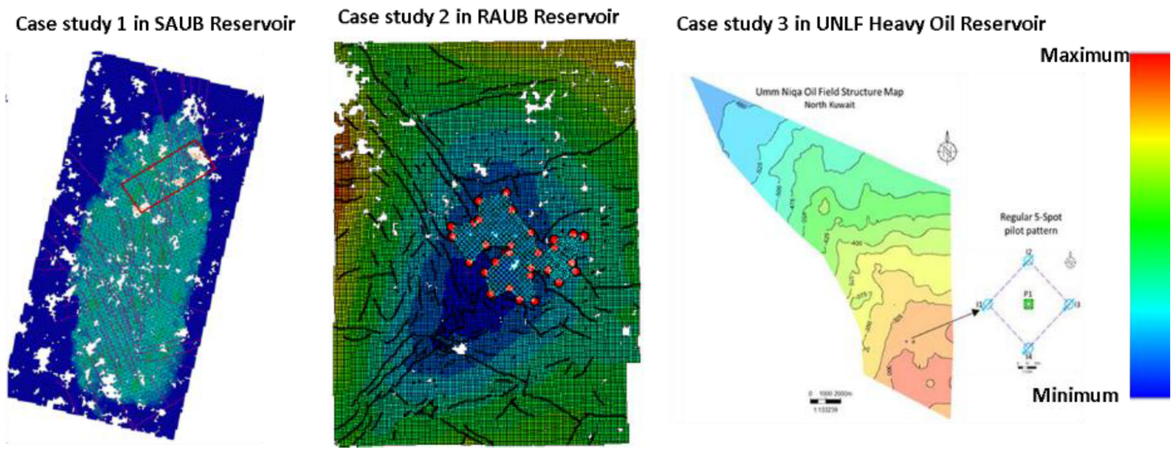

Figure 14—Structural map showing first phase polymer flooding location in North Kuwait reservoirs

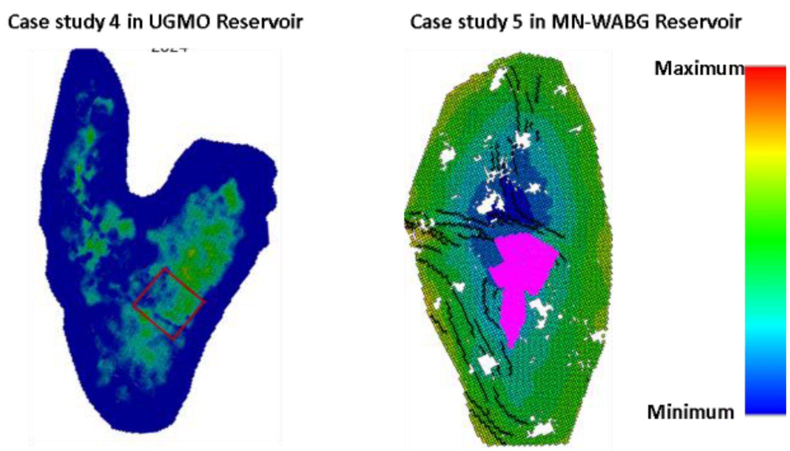

Figure 15—Structural map showing first phase polymer flooding location in West Kuwait reservoirs

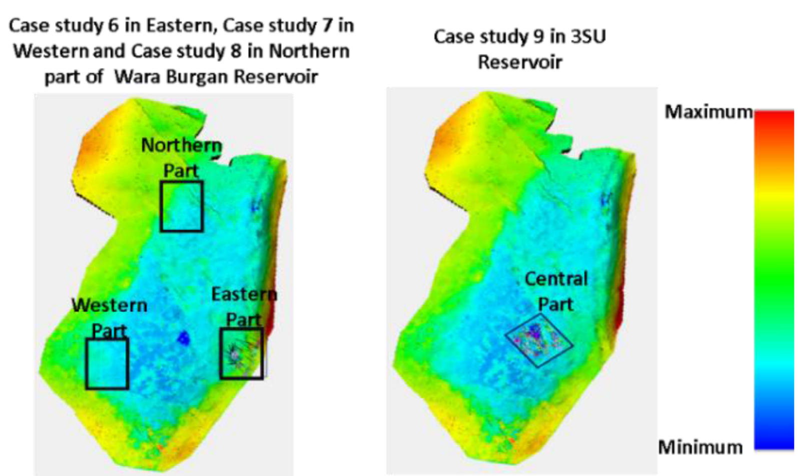

Figure 16—Structural map showing first phase polymer flooding location in South & East Kuwait reservoirs

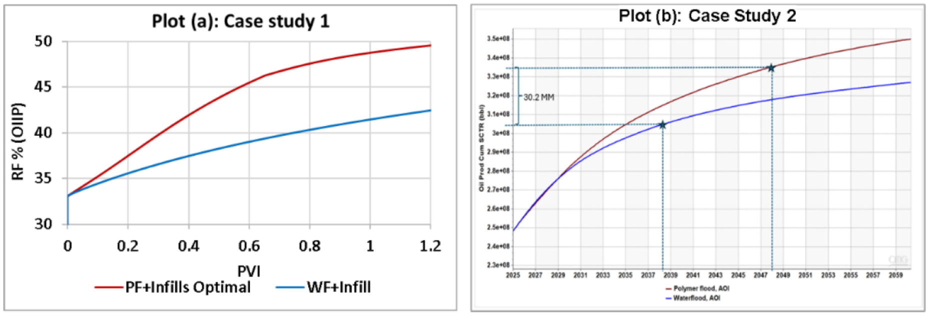

Figure 17—Oil production performance of North Kuwait reservoirs

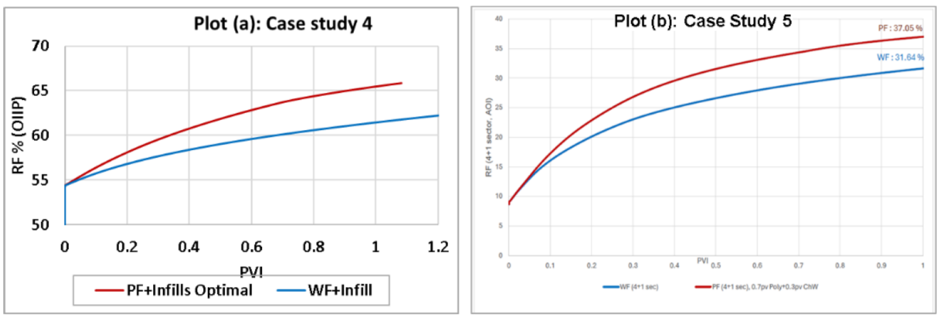

Figure 18—Oil production performance of West Kuwait reservoirs

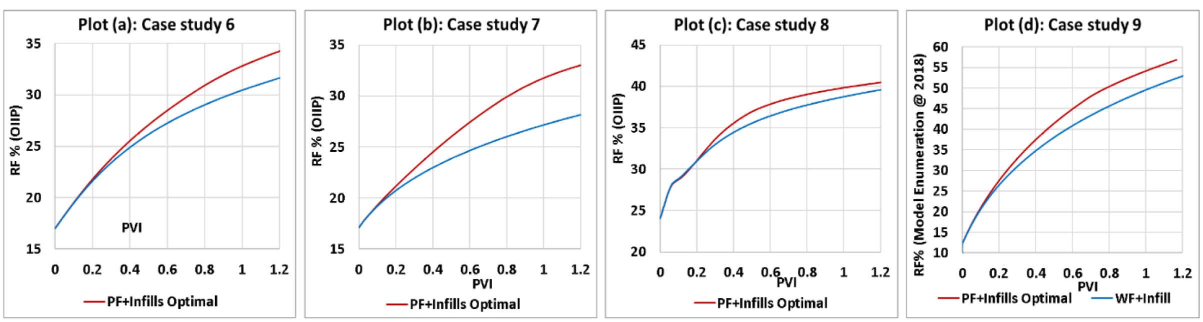

Figure 19—Oil production performance of South and East Kuwait reservoirs

Table 8 summarizes the polymer flooding performance metrics, including total oil produced, incremental oil recovery and unit technical cost (UTC). The results indicate that implementing irregular injection patterns with tighter well spacing—typically ranging from 250 to 400 meters—is crucial for optimizing polymer flood performance. Across the various case studies, the number of injection patterns applied in the first-phase commercial implementation ranged from 5 to 14, depending on reservoir size and complexity. Polymer injection volumes required for effective recovery ranged between 0.5 and 0.8 PV. This injection range yielded substantial incremental oil gains over waterflooding, ranging from 2.8 to 30.2 MMSTB as revealed from different case studies. The corresponding increase in recovery efficiency was found to be between 4.5% and 7.3 % OOIP for most conventional oil reservoirs. Notably, the heavy oil reservoirs demonstrated even higher recovery potential, with polymer flooding contributing to 30% OOIP. Economic evaluations further confirmed the viability of polymer flooding, with unit technical costs (UTC) estimated as 7-23 $/bbl, highlighting the favorable cost-benefit profile of polymer flooding method.

Table 7—LTPIT field results from different case studies from different reservoirs of North, West, Sout and East Kuwait

Case Study	Reservoir	Estimated fracture pressure from step-rate-test (SRT) in psi	Selected polymer name	Polymer description	Polymer retention (µg/g of rock	Injected polymer maximum Polymer concentration (ppm)	Target polymer in-situe viscosity (cp)	Polymer injected in matrix condition	Long term polymer injecvtivity	RRF
Case Study 1	SAUB Reservoir	5020	Flopaam 5115	30 mol % ATBS &9.5-11.3 mD MW	60	1600	3.3	Yes, Pseudo-plastic rheological behavior of injected Polymer i.e., increase estimated in-situ viscosity by increasing polymer concentration.	Good	2.4
Case Study 2	RAUB Reservoir	4266	ZLPAM50830 & ZLPAM40520	50 mol % ATBS & 10-15 mD MW, 15-20 mol % TBS & 8-10 mD MW	52 to 65	1500	3.9		Moderate	2 to 3
Case Study 3	UNLF Heavy Oil Reservoir	120	ZLPAM 40339	25 mol % ATBS & 18 mD MW	24.6	1600	11		Good	
Case Study 4	UGMO Reservoir	4830	ZL 40520	15-20 mol % ATBS & 8-10 mD MW	30	2500	7.5		Good	1.5 to 2
Case Study 5	MN-WABG Reservoir	3209	AN 125 VHM	15 mol % ATBS &10 - 13 mD MW	10 to 12	1500	4.8		Good	2 to 3
Case Study 6	Eastern Wara Burgan reservoir	2635	ZLPAM 40520	15-20 mol % ATBS & 8-10 mD MW	10	1500	5		Good	1.17
Case Study 7	Western Wara Burgan Reservoir	3261				1800	7		Good	2
Case Study 8	Northern Wara Burgan Reservoir	2702				1200	3.85		Moderate	1.8-2.2
Case Study 9	3SU Reservoir	2650			11	1200	3.6		Good	3.5

Table 8—Polymer flooding performance in term of oil produced, incremental oil recovery and UTC

Case Study	Reservoir	Injection patterns	Injection	Incremental oil recovery by polymer flood over water flood (% OOIP)	Incremental cumulative oil by polymer flood over water flood (MMSTB)	UTC ($/bbl)
Case Study 1	SAUB Reservoir	5	0.7 PVI Polymer+0.3 PVI water	7.3	2.8	17.32
Case Study 2	RAUB Reservoir	10	0.8 PVI Polymer+0.2 PVI water	5.43	30.2	21.26
Case Study 3	UNLF Heavy Oil Reservoir	1 (inverted 5-spot)	0.8 PVI Polymer+0.2 PVI water	30	42.8	
Case Study 4	UGMO Reservoir	13	0.8 PVI Polymer+0.2 PVI water	4.5	16.94	20.5
Case Study 5	MN-WABG Reservoir	5	0.7 PVI Polymer+0.3 PVI water	5.4	21.4	15.3
Case Study 6	Eastern Wara Burgan reservoir	9	0.8 PVI Polymer+0.2 PVI water	2.4	6.396	14.81
Case Study 7	Western Wara Burgan Reservoir	10	0.8 PVI Polymer+0.2 PVI water	4.7	6.25	23.2
Case Study 8	Northern Wara Burgan Reservoir	6	0.5 PVI Polymer+0.5 PVI water	1.7	1.3	10
Case Study 9	3SU Reservoir	14	0.2 PVI Water + 0.7PVI Polymer + 0.1PVI Water	4.9	13.3	7.37

Conclusions

- Laboratory investigations and core flooding experiments confirmed that synthetic polymers containing 15–50% ATBS molecules with molecular weights 8-18 million Daltons exhibit strong mechanical, chemical, and thermal stability under challenging reservoir conditions, including high salinity (up to 270,000 ppm TDS), elevated hardness (up to 40,000 ppm), and temperatures up to 210°F. These polymers maintained effective viscosities within the target range of 3.3–7 cP at economically viable concentrations (1200–2500 ppm), while demonstrating low dynamic retention (10–65 µg/g-rock) and acceptable resistance factors (1.1–3.5).
- Long-term polymer injectivity tests (LTPIT) successfully validated polymer performance under field-scale conditions, maintaining consistent feedwater quality (TSS<5ppm, Oil in Water<10ppm, Dissolved Oxygen<10ppb, Iron content<1.5ppm, H2S<5ppm and Bacteria<10^3 counts/ml). No polymer breakthrough was observed in adjacent production wells within a 200–300 m radius over a 3–6 month period, indicating efficient polymer propagation. Polymer injectivity tests also confirmed the polymer's pseudoplastic (shear-thinning) behavior near the wellbore under matrix flow conditions, with in-situ viscosity increasing alongside polymer concentration and improving injection profiles post-treatment.
- Field deployment further demonstrated the technical feasibility and economic viability of phase commercial polymer flooding in reservoir harsh conditions and using high TDS produced water as injection source. Optimizing injection patterns and utilizing existing infrastructure with tighter

well spacing (200–400 m) led to efficient reservoir sweep and enhanced oil recovery. Optimized polymer flooding required only 0.5–0.8 PVI polymer to achieve an incremental recovery of 4.5–7.3% OOIP, resulting in additional oil production 2.8–30.2 (MMSTB across various case studies in Kuwait's sandstone and carbonate reservoirs. Economic analysis showed that polymer flooding can be implemented at unit technical costs between $7 and $23 per barrel, highlighting its cost-effectiveness and potential for wider application in similar reservoirs.

Acknowledgments

The authors would like to extend their sincere gratitude to Kuwait Oil Company (KOC) for granting permission to publish this paper. Finally, the authors acknowledge the entire multidisciplinary team whose coordinated efforts made this work possible—from planning to pilot implementation.

Nomenclatures / Acronyms

Symbol	***Description***
TDS	*Total dissolved solids*
LTPIT	*Long term polymer injectivity test*
RF	*Resistance factor*
RRF	*Residual resistance factor*
UTC	*Unit technical cost*
NPV	*Net present value*
EOR	*Enhanced oil recovery*
HPAM	*Particularly hydrolyzed polyacrylamide*
ATBS	*Acrylamide-tertiary-butyl-sulfonic-acid*
SRT	*Step rate test*
IOR	*Improved oil recovery*
HPT, SNL and ILT	*High-Precision Temperature, Spectral Noise Logging and Injection Logging Test, respectively*
BHP	*Bottom hole pressure*
PFO	*pressure fall off*
WHP	*Well head pressure*
Ppm	*part per million*
OOIP	*Original oil in place*
MMSTB	*Million stock tank barrels*
PVI	*Pore volume injection*
H, Sw,end HM and k	*Reservoir thickness, current water saturation and permeability, respectively*
KH	*permeability-thickness*

References

Abdullah, M. B., Delshad, M., Sepehrnoori, K., Balhoff, M. T., Foster, J. T., & Al-Murayri, M. T. (2023). Physics-based and data-driven polymer rheology model. *SPE Journal*, **28**(04), 1857–1879.

Abduraheem, H., Hassan, H., Al-Ajmi, N., Khan, M. Y., Al-Khuzam, S., Kakade, S., Banerjee, A., Balushi, A., Boareki, N., Andrianov, A., (2025, November). First Long-term Polymer Injectivity Pilot Targeting The Raub Reservoir In North Kuwait: Successful Implementation And Evaluation Of Two Polymers Using High-salinity Effluent Water. In Abu Dhabi International Petroleum Exhibition and Conference. SPE-229950-MS.

AlAbdullah, M. B., Algharaib, M., Seright, R. S., & Sanaseeri, A. (2025, May). A Multi-Scale Evaluation of Polymer Injectivity and Fracture Behavior in the Burgan Field. In SPE EOR Conference at Oil and Gas West Asia (p. D021S013R001). SPE.

Al-Mayyan, H. E. A., Al-Dabbous, A., Eadulapally, S., Al-Rukaibi, D., Al-Abbas, R., Al-Murayri, M. T., … & Delamaide, E. (2025, May). Optimizing Polymer EOR Performance Under High Salinity and High Temperature Conditions for a Clastic Reservoir in Kuwait. In SPE EOR Conference at Oil and Gas West Asia (p. D021S019R004). SPE.

Al-Mayyan, H. E., Al-Kharji, A., Khan, M. Y., Al-Subaihi, M. A., Abdul-Aziz, W. J., Mohamed, M. F., … & Andrianov, A. (2024, November). Polymer Flooding in a Major Carbonate Reservoir Using High-Salinity Water: Promising Pilot Results and Techno Economic Evaluation for the Umm Gudair Minagish Oolite Reservoir in West Kuwait. In Abu Dhabi International Petroleum Exhibition and Conference (p. D031S081R003). SPE.

Al-Murayri, M. T., Abdulraheem, H. F., Qureshi, F. A., Hassan, A. A., Eadulapally, S. K., Garcia, J. G., … & Nesbit, J. (2023, October). Significant Oil Uplift from Field Implementation of the First High Salinity Polymer-Flood Targeting a Sour Heavy Oil Reservoir Worldwide. In Abu Dhabi International Petroleum Exhibition and Conference (p. D031S103R004). SPE.

Al-Murayri, M. T., Al-Mayyan, H. E., Al-Ghnemi, M. F., Pitts, M., Dean, E., Wyatt, K., … & Skeans, E. (2019, March). Sabriyah Lower Burgan Field EOR Development Study: Use of Sacrificial Wells to Minimize Water Encroachment. In SPE Middle East Oil and Gas Show and Conference (p. D032S101R002). SPE.

Al-Murayri, M. T., Al-Mayyan, H. E., Faraj, A. A., AlAbdullah, M. B., Pitts, M., & Wyatt, K. (2017, May). Evaluation of Enhanced Oil Recovery Technologies for the Sabriyah Lower Burgan Reservoir Kuwait. In SPE Reservoir Characterisation and Simulation Conference and Exhibition (p. D021S008R003). SPE.

Al-Murayri, M. T., Al-Mayyan, H. E., Moudi, K., Al-Ajmi, F., Pitts, D., Wyatt, M. J., … & Dean, E. (2018, March). Chemical EOR economic evaluation in a low oil price environment: Sabriyah lower burgan reservoir case study. In SPE EOR Conference at Oil and Gas West Asia (p. D021S008R003). SPE.

Al-Murayri, M. T., Al-Qattan, A., Hassan, A. A., Al-Ghanim, W., Jadhav, P., Kakade, S., … & Dean, E. S. (2022, October). First Polymer Injectivity Test in the Wara Formation of the Great Burgan Field: Strategic Milestone to Fast-Track Phased Commercial Polymer-Flooding Development. In Abu Dhabi International Petroleum Exhibition and Conference (p. D031S111R002). SPE.

Al-Murayri, M. T., Al-Qattan, A., Wafaa, A. G., Kakade, S., Banerjee, A., Mascagnini, C., … & Malcolm, P. (2023, March). Integrating High Resolution Sector Scale Dynamic Modeling into a Successful Long Term Pilot Polymer Injectivity Test for the Wara Reservoir of the Greater Burgan Field of Kuwait. In SPE Middle East Oil and Gas Show and Conference (p. D011S086R002). SPE.

Al-Murayri, M. T., Hassan, A. A., Alajmi, N. A., Khan, M. Y., Al-Qattan, A., Jamal, M., … & Dean, E. (2023, October). First Successful Long-Term Polymer Injectivity Test in the Upper Burgan Formation of the Sabriyah Field to Fast-Track Phased Commercial Polymer-Flooding Development. In Abu Dhabi International Petroleum Exhibition and Conference (p. D031S077R003). SPE.

Al-Murayri, M. T., Hassan, A., Alajmi, N., Andrianov, A., Garcia, J., de Kruijf, A., … & Li, E. (2024, April). Practical Appraisal of Sulfonated Polymers for a Sour High Salinity and Heavy Oil Reservoir in Kuwait to Fast-Track Field Implementation. In SPE EOR Conference at Oil and Gas West Asia (p. D011S007R003). SPE.

Al-Murayri, M. T., Kamal, D. S., Al-Sabah, H. M., AbdulSalam, T., Al-Shamali, A., Quttainah, R., … & Dean, R. M. (2019, October). Low-Salinity polymer flooding in a high-temperature low-permeability carbonate reservoir in West Kuwait. In SPE Kuwait Oil and Gas Show and Conference (p. D043S022R004). SPE.

Al-Murayri, M. T., Kamal, D. S., Garcia, J. G., Delshad, M., Britton, C., & Fortenberry, R. (2018, December). Geochemical Modeling to Evaluate the Performance of Polymer Flooding in a Highly Sour Sandstone Heavy Oil Reservoir. In SPE International Heavy Oil Conference and Exhibition (p. D011S002R001). SPE.

Al-Murayri, M. T., Kamal, D., Al-Mahmeed, N., Al Kharji, A., Baroon, H., Pitts, M. J., … & Obernyer, S. (2021, December). Design of an Economical Polymer Flood Pilot Addressing Field Development Uncertainties in the Sabriyah Upper Burgan Reservoir of North Kuwait. In Abu Dhabi International Petroleum Exhibition and Conference (p. D022S180R001). SPE.

Al-Murayri, M. T., Kamal, D., De Kruijf, A., & Andrianov, A. (2023, October). Impact of Produced EOR Polymer on Production Facilities. In Abu Dhabi International Petroleum Exhibition and Conference (p. D031S077R005). SPE.

Al-Murayri, M. T., Kamal, D., de Kruijf, A., Imqam, A., Andrianov, A., Liu, E., & Al-Rawahi, S. (2024, April). Improved Water Management Through the Usage of Sulfonated Polymers. In SPE EOR Conference at Oil and Gas West Asia (p. D031S028R001). SPE.

Al-Qattan, A., Al-Murayri, M., Almahmid, N., Alkhuzam, S., Sanaseeri, A., Alajmi, N., … & Adrianov, A. (2024, February). First Long-Term Polymer Injectivity (LTPI) Test in Magwa Field Targeting Medium Quality Formation in Wara, Great Burgan Field. In International Petroleum Technology Conference (p. D011S016R004). IPTC.

Dupuis, G., Antignard, S., Giovannetti, B. et al 2017. A New Thermally Stable Synthetic Polymer for Harsh Conditions of Middle East Reservoirs. Part I. Thermal Stability and Injection in Carbonate Cores. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 13–16 November. SPE- 188479- MS. https://doi.org/10.2118/188479-MS.

Dupuis, G., Metidji, M. O., Terra, N., Nesbit, J., Mishra, A., Kamal, D., & Al-Murayri, M. T. (2024, April). First Field Implementation of a Universal Method to Determine Polymer Concentration in Production Fluids During Polymer Flooding Operations Using Size Exclusion Chromatography. In SPE Improved Oil Recovery Conference? (p. D041S029R003). SPE.

Eadulapally, S. K., Taha, A. M. M., Khan, M. Y., Ali, A. M. H. E., Fahad, A. M., & Deema, A. (2024, November). Unraveling Geological Complexity and Implementing Risk Mitigation Strategies in EOR Pilots Within Carbonate Reservoirs. In Abu Dhabi International Petroleum Exhibition and Conference (p. D011S022R005). SPE.

Hassan, A. M., Al- Shalabi, E. W., and Ayoub, M. A. 2022. Updated Perceptions on Polymer- Based Enhanced Oil Recovery toward High- Temperature High- Salinity Tolerance for Successful Field Applications in Carbonate Reservoirs. *Polymers (Basel)* **14** (10). https://doi.org/10.3390/polym14102001.

Khan, M. Y., Al-Murayri, M. T., Eadulapally, S. K., Al-Mayyan, H. E., Alrukaibi, D., & Al-Kharji, A. (2024, February). Strategies to Design Fit for Purpose EOR Pilots by Integrating Dynamic and Static Data in a Highly Heterogeneous Oolitic Carbonate Reservoir. In International Petroleum Technology Conference (p. D021S059R010). IPTC.

Khan, M. Y., Kamal, D., Al-Mayyan, H. E. A., Pradhan, A. K., Eadulapally, S. K., & Al-Murayri, M. T. (2023, March). Transforming Challenges into Opportunities to Design Fit for Purpose Long Term Polymer Injectivity (LTPI) Pilots in Umm Gudair Minagish Oolite Reservoir, West Kuwait. In SPE Middle East Oil and Gas Show and Conference (p. D021S086R007). SPE.

Lake, L., Johns, R. T., Rossen, W. R., Pope, G. A. (2014). Fundamentals of Enhanced Oil Recovery. *Society of Petroleum Engineers*. doi.org/10.2118/9781613993286

Levitt D. B., Pope G. A., "Selection of Screening of Polymer for Enhanced Oil Recovery". SPE Improved Oil Recovery Symposium, 20-23 April, Tulsa, Oklahoma. SPE 113845. 2008

Seright, R.S., Wavrik, K.E., Zhang, G., AlSofi, A.M., 2021. Stability and behavior in carbonate cores for new EOR polymers at elevated temperatures in hard saline brines. *SPE Reservoir Eval. Eng*. **24** (1),1e18. https://doi.org/10.2118/200324-PA.

Sorbie, K. S. (1991). Polymer-Improved Oil Recovery. Boca Raton, Fl: CRC Press.

Standnes, I. Skjevrak. (2014, August 23). Literature review of implemented polymer field projects. *Journal of Petroleum Science and Engineering*. **122** (2014) 761–775.

Yunus, K. M., Dawood, K., Ali, A. M. H. E., Kumar, P. A., Kumar, E. S., & Taha, A. M. M. (2023). Transforming Challenges into Opportunities to Design Fit for Purpose Long Term Polymer Injectivity (LTPI) Pilots in Umm Gudair Minagish Oolite Reservoir, West Kuwait.